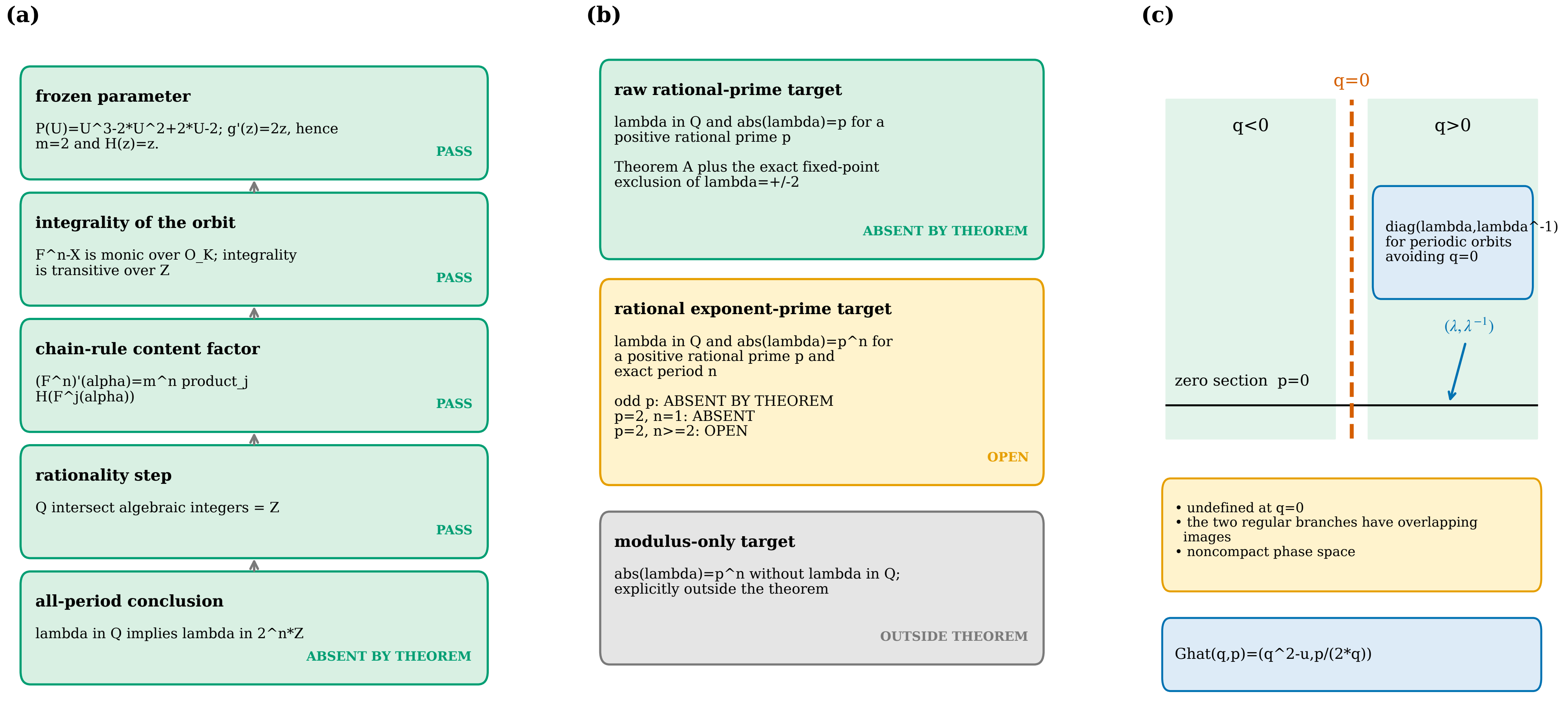
(a)
frozen parameter
P(U)=U^3-2*U^2+2*U-2; g'(z)=2z, hence m=2 and H(z)=z.
PASS
integrality of the orbit
F^n-X is monic over O_K; integrality is transitive over Z
PASS
chain-rule content factor
(F^n)'(alpha)=m^n product_j H(F^j(alpha))
PASS
rationality step
Q intersect algebraic integers = Z
PASS
all-period conclusion
lambda in Q implies lambda in 2^n*Z
ABSENT BY THEOREM
(b)
raw rational-prime target
lambda in Q and abs(lambda)=p for a positive rational prime p
Theorem A plus the exact fixed-point exclusion of lambda=+/-2
ABSENT BY THEOREM
rational exponent-prime target
lambda in Q and abs(lambda)=p^n for a positive rational prime p and exact period n
odd p: ABSENT BY THEOREM
p=2, n=1: ABSENT
p=2, n>=2: OPEN
OPEN
modulus-only target
abs(lambda)=p^n without lambda in Q; explicitly outside the theorem
OUTSIDE THEOREM
(c)
q=0
q<0
q>0
diag(lambda,lambda^-1) for periodic orbits avoiding q=0
$(\lambda, \lambda^{-1})$
zero section p=0
• undefined at q=0
• the two regular branches have overlapping images
• noncompact phase space
Ghat(q,p)=(q^2-u,p/(2*q))